

CASE 2

**BY:
SHRUTHI
REN REN
ADITHYA
XIAOER
SREELAKSHANA
SHUSHMA SADANAND**



Context:

A BLS on YouTube tries to measure awareness. Awareness is correlated with age and purchase history.

Survey data

Group	age_segment	respondents	yes_awareness	variance (v)
Exposed	18-34	1000	420	0.2436
Control	18-34	1000	380	0.2356
Exposed	35-54	1000	460	0.2484
Control	35-54	1000	410	0.2419

01

Compute the awareness lift and significance test for one segment (your choice)

Lets Take :

Group	Age_Segment	Respondents (r)	Yes_awareness (y)	Variance (v)
Exposed	18-34	1000 (r1)	420 (y1)	0.2436
Control	18-34	1000 (r2)	380 (y2)	0.2356

$$\begin{aligned}P1 &= y1/r1 \\&= 420/1000 \\&= 0.420\end{aligned}$$

$$\begin{aligned}P2 &= y2/r2 \\&= 380/1000 \\&= 0.380\end{aligned}$$

Awareness Lift:

$$\begin{aligned}&(\text{Absolute lift}) \\&\Delta = P1 - P2 \\&= 0.420 - 0.380 \\&= 0.04\text{pp} \\&(\text{4 percentage points})\end{aligned}$$

$$\begin{aligned}&(\text{Relative lift}) \\&= (P1 - P2) / P2 \\&= (0.42 - 0.38) / 0.38 \\&= 0.04 / 0.38 \\&= 0.105 \\&= 10.5\%\end{aligned}$$

Significance Test (Z test):

$$p = \frac{(y_1 + y_2)}{(r_1 + r_2)}$$

$$\begin{aligned} p &= (420 + 380) / (1000 + 1000) \\ &= 800 / 2000 \\ &= 0.40 \end{aligned}$$

$$Z = \frac{P_1 - P_2}{\sqrt{P(1 - P) \left(\frac{1}{r_1} + \frac{1}{r_2} \right)}}$$

$$\begin{aligned} 1/y_1 &= 1/1000 \\ 1/y_2 &= 1/1000 \\ (1/r_1) + (1/r_2) &= 2/1000 \\ &= 0.002 \end{aligned}$$

$$\begin{aligned} Z &= 0.42 - 0.38 / \text{sqrt} (0.4 (1 - 0.4) (0.002)) \\ &= 0.04 / \text{sqrt} (0.4 \times 0.6 \times 0.002) \\ &= 0.04 / \text{sqrt} (0.00048) \\ &= 0.04 / 0.0219 \\ Z &= 1.83 \end{aligned}$$

02

Decide: should the advertiser report a single overall lift or stratify by groups?
Justify with pros and cons.

Stratify by Groups

Pros:

- Reveals performance differences across demographics.
- Enables targeted optimization (e.g., age-specific creative or media buys).
- More transparent and actionable for future campaigns.

Cons:

- More complex to communicate.
- May dilute the perceived impact if some segments show weak or no lift.

Single Overall Lift

Pros:

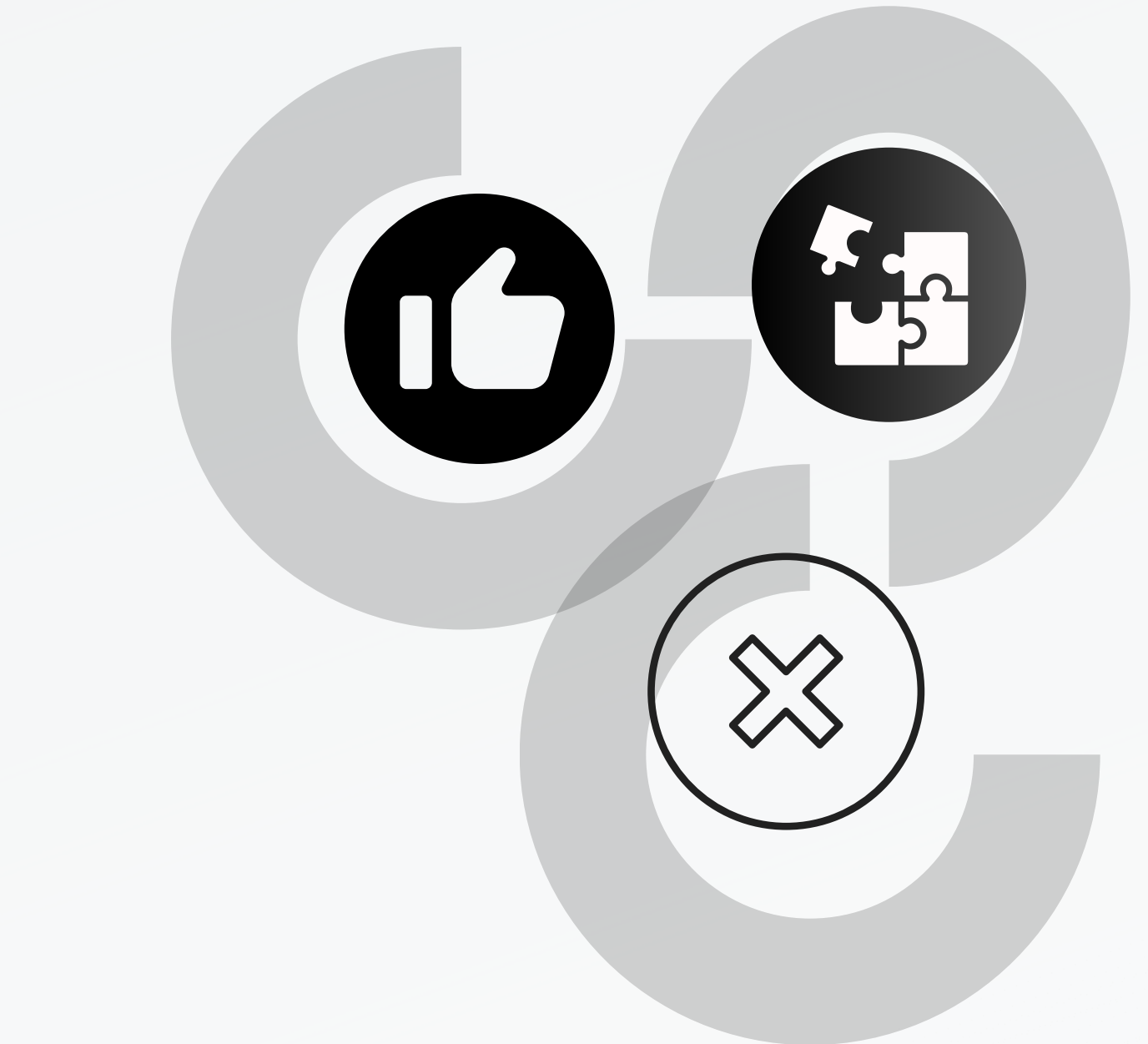
- Simpler narrative for stakeholders.
- Easier to benchmark against other platforms.

Cons:

- Masks underperformance in key segments.
- Risks misallocating budget if lift is driven by one group only.

Decision :

Stratify.- The lift varies by age, and strategic decisions depend on knowing where YouTube performs best.



03

Imagine you are the agency: the client asks “What’s the next best step to understand if YouTube really works?” You can propose any design you like.

MAIN IDEAS



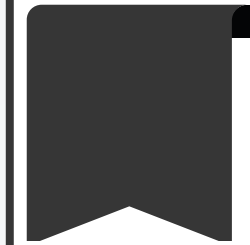
Expand the sample size

generate more accurate and convincing results



Improve the randomization design

purchase history
propensity score
more balanced Treatment and Control groups



Shift the focus to core conversion rates

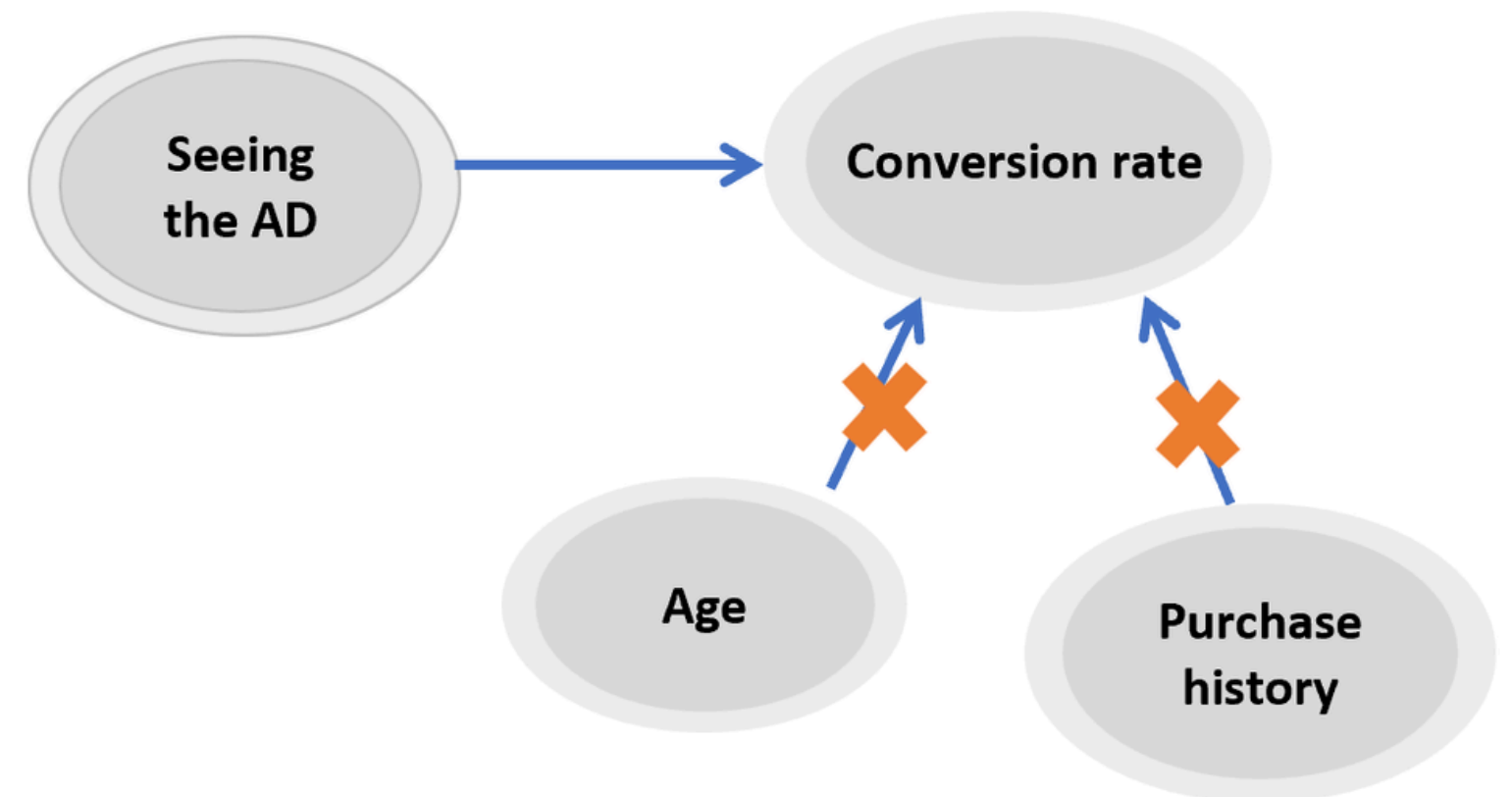
Purchase Rate directly reflects the business impact

03

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STEP 1 Design Overview

- Sample size: 20,000 users
- Randomization: Generate propensity score based on age + purchase history, stratify/match then randomize
- Treatment: Exposed to YouTube ads
- Control: No ads shown
- Split ratio: 80% treatment / 20% control



03

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STEP 2 Data Collection

- User profile:** age, gender, region.
- History:** past 6 months frequency & spend.
- During experiment:**
 - Ad exposure record (seen ad: 0/1)
 - Experiment outcome: whether purchased, purchase amount

user_id	age_segment	hist_6m_purchase_freq	hist_6m_purchase_amount	exposed	purchased	purchase_amount	gender	region
1	18-34	0	0	1	1	60	1	A
2	35-54	0	0	1	1	50	0	B
3	35-54	3	30.912	0	1	120	1	B
4	35-54	1	8.768	1	0	0	1	A
5	18-34	0	0	0	0	0	0	A
6	18-34	0	0	0	1	80	1	B

03

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STEP 3 KPI Framework

- **Primary KPI:** $\text{Purchase rate} = \text{buyers} / \text{total users}$
- **Secondary KPIs:**
 - **AOV**(Average order value) $= \text{Total purchase amount} / \# \text{ of purchasers}$
 - **Total sales uplift** $= \text{Sales in Treatment} - \text{Sales in Control}$
 - **iROAS** $= \text{Incremental sales uplift} / \text{Ad spend}$

03

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STEP 4 Analysis

		Examples		
Metric	Difference / Uplift	Treatment Group	Control Group	Lift
Purchase Rate Lift	$\Delta = \text{PurchaseRate.T} - \text{PurchaseRate.C}$	10% (100/1000)	5% (50/1000)	5%
Statistical Significance (z-test)	$z = \frac{p_T - p_C}{\sqrt{p(1-p) \left(\frac{1}{n_T} + \frac{1}{n_C} \right)}}$	-	-	4.1 Z>K(1.96), significant different Z<K(1.96), not significant different
Segmentation	break down by age group	-	-	18–34 yrs: +8%, z = 2.8, (significant) 35–54 yrs: +4%, z = 1.3, (not significant)
Sales Uplift	$\text{SalesUplift} = \Delta \times N.\text{Treatment} \times \text{AOV}$	-	-	+€10,000
iROAS	$\text{iROAS} = \text{SalesUplift} / \text{Ad Spend}$	-	-	2

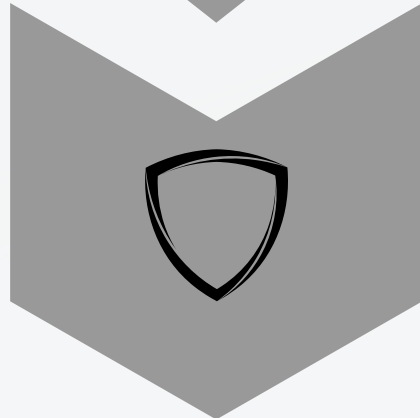
04 Recommendations

A Balanced Investment Strategy



Prioritize 35–54 with Incremental Budget

Allocate the majority of new or incremental budget to this segment, capitalizing on its proven effectiveness and strong ROI potential.



Maintain Baseline Investment in 18–34

Ensure continued engagement with this younger demographic to foster long-term brand equity and loyalty.

Our suggested split is approximately **~65% for 35–54** and **~35% for 18–34**. This allows us to capture short-term gains while simultaneously nurturing future consumers for sustainable success.

CASE - 2

